

The acute effects of medium-chain triglyceride supplementation on metabolic rate and fat oxidation at rest in healthy young adults

Abstract:

Introduction:

Medium-chain triglycerides (MCT) are rapidly absorbed by the body and metabolised for energy, yet existing evidence for their effects on resting metabolic rate (RMR) and fat oxidation in recreationally active young adults are limited and inconsistent. This study aims to investigate the acute effects of MCT supplementation on RMR and fat oxidation at rest.

Methods:

A total of nine healthy young adults (5 male and 4 females; aged 21.4 ± 0.7 years old; BMI $23.3 \pm 3.9 \text{ kg/m}^2$) took part in before and after intervention study. Breath-by-breath gas analysis was done to measure RMR and substrate utilisation. Conditions were standardised by testing participants in a fasted state, having avoided alcohol, caffeine and exercise the day prior. After initial baseline testing, the participants consumed a drink containing 30g of MCT oil, 100ml of lactose-free low-fat milk and a teaspoon of cacao powder. They returned after 3 hours to repeat the testing.

Results:

Acute MCT supplementation led to a significant decrease in RER suggesting a shift towards fat oxidation at rest. Pre-supplementation values: $0.904 [0.876 - 0.933]$; post-supplementation values: $0.882 [0.863 - 0.894]$ ($W=36.00, p=0.008, r=1.000$). RMR however did not show a notable change after MCT consumption. Pre-supplementation values: $2408 [1783 - 3181] \text{ kcal/day}$; post-supplementation: $2430 [1902 - 3239] \text{ kcal/day}$ ($W=15.00, p=0.742, r=-0.167$). Other secondary outcome measures did not show any changes ($P > 0.05$).

Conclusion:

The findings partially support the hypothesis. Acute ingestion of 30g of C8 MCT oil has significantly reduced RER at rest in young healthy adults indicating a greater usage of fat oxidation for energy. However, RMR shows negligible changes pre and post ingestion. A larger sample size is needed with a control condition in order to confirm these changes.

Key terms:

Medium-chain triglycerides; respiratory exchange ratio; resting metabolic rate; fat oxidation; indirect calorimetry.

Introduction:

Medium-chain triglycerides (MCTs) are made up of fatty acid chains consisting of 6 to 12 carbons. Typically, these make up a small proportion of one's diet and can be found in foods including coconut oil and ghee[1]. Capric(C10) and Caprylic(C8) are the most active forms of MCTs and are more commonly studied. In contrast, long-chain triglycerides (LCTs) are the most common type of dietary lipids found in the diet containing more than 14 carbons. As a result of the differences in chain lengths, MCTs are

digested, absorbed and metabolised differently compared to LCTs, giving it different physiological properties[2].

LCTs are normally emulsified by bile salts and incorporated into micelles before they can be absorbed. MCTs on the other hand are hydrolysed much quicker and do not require bile[3]. LCTs are initially absorbed into the lymphatic system and packaged as chylomicrons. However, MCTs are directly then absorbed into the portal circulation where they reach the liver. This pathway allows MCTs to reach the liver quicker and in high concentrations where they can undergo fatty acid oxidation to release energy and produce ketone bodies, which can be used as an alternative fuel source for peripheral tissues including the brain and skeletal muscle[4].

Once they enter the hepatocyte, the oxidation physiology is different for MCTs and LCTs. LCTs require a carnitine shuttle to cross the inner mitochondrial membrane before they can be oxidised. Whereas MCTs can cross the membrane without the need for a carnitine shuttle which allows for a faster route of entry and rapid oxidation[4]. These characteristics (rapid digestion, absorption and oxidation) theorises MCT is a possible supplementation for increasing energy production and fat utilisation.

Resting metabolic rate (RMR) is the amount of energy used by a person at rest to ensure essential physiology such as maintaining blood flow, breathing, thermoregulation and homeostasis. It makes up 60-70% of an individual's total daily energy expenditure (TDEE)[5]. As a result, even a small increase in RMR could lead to meaningful changes in weight regulation over time.

There are many factors that determine RMR. The most significant one being fat-free mass due to the high metabolic activity of lean tissue compared to adipose tissue which is relatively metabolically inactive. Other factors include age and sex[6]. Throughout early adulthood, RMR remains relatively stable, however, there is quite a substantial variation between individuals. This is especially the case between males and females due to differences in body composition[7].

RMR is typically measured using indirect calorimetry and breath-by-breath analysis. This is the gold standard method in a laboratory setting. Measurements such as oxygen consumption (VO_2) and carbon dioxide production (VCO_2) are used to calculate an estimate of energy expenditure[8].

Substrate oxidation refers to the input of carbohydrate and fat oxidation towards energy production. This can be examined through the respiratory exchange ratio (RER) which refers to the ratio of carbon dioxide production in relation to oxygen consumption. An RER closer to 0.7 indicates predominant fat oxidation. This is normally seen in a fasted state where insulin availability is low or during rest where the demand for energy is little. RER of 1.00 reflects mainly carbohydrate utilisation. Values above 1.00 reflect non-steady state conditions such as lactate accumulation resulting from metabolic acidosis[9].

Breath-by-breath gas analysis using the metalyzer 3b allows for continuous measurement of these metabolic variables. The testing is done under standardised conditions. This involves fasting overnight, avoiding intense exercise prior to testing, avoiding caffeine, alcohol and nicotine[10].

Several studies have demonstrated MCT consumption increases RMR compared to LCT. St-Onge & Jones(2002) presents a significant increase in postprandial energy expenditure following the consumption of MCT-rich meals compared to LCT in healthy adults measured via indirect calorimetry over a 5.5-hour postprandial period[11]. Some studies have also reported an increased reliance of fat oxidation for energy

after MCT consumption. Alexandrou(2007) reported notable increase in fat oxidation and decrease in carbohydrate usage in young healthy adults following MCT consumption[12]. However, across all the existing studies in regard to acute MCT consumption, the findings are not consistent. Some have reported no significant changes in resting metabolic rate or fat oxidation.

Even with increasing evidence, there are some important gaps in the literature. Most of the studies were conducted in older populations outside the healthy BMI range. There are limited studies involving healthy young adults that are recreationally active. Additionally, the designs of the studies are varied and inconsistent including differences in MCT dose, timing and method of administration. This limits direct comparison between studies and indicates the need for a controlled study in younger, healthy adults using gold standard equipment for measuring metabolic variables and standardised protocols.

Therefore, the aim of this study is to investigate the acute effects of a 30g dose of MCT oil on resting metabolic rate (RMR) and fat oxidation at rest in healthy young adults, measured three hours post-ingesting using breath-by-breath gas analysis. With the physiology of MCT in mind, the hypothesis is that the supplement would increase RMR and reduce RER compared to the baseline values prior to taking the supplement.

Methods:

Design

In order to investigate the effects of acute MCT supplementation on resting metabolic rate and fat oxidation at rest, the study followed a within-subjects pre-post intervention design. Having each participant be their own control allows for increased reliability and validity by reducing inter-individual variability[13]. The testing for all participants was done at the Human Performance Lab at Queen Mary University. Prior to recruitment and testing, ethical approval was granted by the QMEthics committee (DSREC; Review Reference: 2026-2073-3177, Project ID:2073). Before testing commenced, the volunteers were given a participant information sheet explaining the purpose of the study and why they were chosen and what is expected of them. They were also given a consent form to sign prior to testing.

Participants

A total of nine healthy young adults were recruited for the study through poster advertisement. Participants had to meet the eligibility criteria in order to include in the study. Table 1 outlines the inclusion and exclusion criteria. Participant anthropometric characteristics are presented in Table 2.

Table – Inclusion and exclusion criteria for participant recruitment

Inclusion Criteria	Exclusion Criteria
Aged 18-30 years old	Metabolic disorder
Recreationally active	Cardiovascular disease
Healthy, no chronic illness	Current medication
Non-smoker	Pregnant or breastfeeding

	Allergy to MCT/Milk
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Table – Anthropometric characteristics of recruited participants (n = 9)

Descriptives				
	Age	Weight_kg	BMI	Height_cm
N	9	9	9	9
Mean	21.4	69.2	23.3	172
Standard deviation	0.726	15.3	3.90	8.06
Minimum	20	49.7	19.2	157
Maximum	22	89.5	29.6	183

Pre-test standardisation

Before attending the session for testing, the participants were asked to arrive fasted overnight, without having taken any alcohol or caffeine. They were asked to avoid strenuous exercise for 24 hours prior as it can influence RMR[14]. Once they arrived at the lab, a nutritional questionnaire was completed to ensure their dietary compliance and that they have adhered to the standardisation requirements. The consent form is then provided to ensure the participants know the risks involved. Anthropometric measurements including weight, height, age and sex were recorded prior to testing.

Protocol

The resting metabolic measurements were conducted using a Cortex Metalyzer 3B breath-by-breath gas analyser in a laboratory at room temperature (21oC). This device has been proved to provide valid and reliable assessment of gas exchange and energy expenditure at rest[15]. The equipment was calibrated using reference gases of known concentration. Participants were fitted with a face mask and seated in a comfortable supine resting position for 45 minutes. They were also fitted with a Polar heart rate monitor to measure heart rate.

The first 15 minutes were not included in the data analysis as it would take some time for the participants to acclimatise to the laboratory setting and for the metabolic variables to stabilise[16]. The mean values were calculated using the final 30 minutes of the test. The participants were instructed to remain calm, avoid talking and making unnecessary movements in order to achieve a rested state. The conditions for pre-supplement test were done between 8:00am and 10:00am while the post-supplement test was carried out between 11:00am and 1:00pm. Apart from this, the conditions were identical

Intervention

After the initial testing, the participants consumed a drink made with 30g of Ketosource Pure C8 MCT oil (caprylic acid - coconut oil derived), 100ml of heated lactose-free fat-free milk (0.5% fat; 0.5g fat, 3.4g protein, 4.8g carbohydrate) and 3g of cacao powder (0.8g fat, 0.8g protein, 0.8g carbohydrate), totalling approximately 280 kcal. The drink was prepared by the researcher in a standardised manner using accurate measurement tools. After consumption, participants left the laboratory for three-hours during which they were asked to avoid food and physical activity. They were not supervised during this period and adherence

to the instructions could not be verified. They would return to the same laboratory three hours post-ingestion to repeat metabolic testing. This three-hour period was decided based on evidence showing peak MCT response and postprandial energy expenditure occur in this timeframe[17][18].

Outcome measures

Several outcome measures were examined in the study. The primary outcome measures were VO_2 and VCO_2 which were needed to calculate resting metabolic rate. RMR for each breath was calculated using the Weir-Equation: $\text{RMR}(\text{Kcal}/\text{min}) = (3.941 \times \text{VO}_2) + (1.106 \times \text{VCO}_2)$ [19]. This was then multiplied by 1440 to give the RMR per day[19]. Another primary measure was the respiratory exchange ratio (RER) as it indicates substrate utilisation in regard to carbohydrate and fat[20]. Other outcomes were also examined as secondary measures including heart rate(HR), breathing-frequency(BF), relative oxygen-intake(VO_2/kg), oxygen pulse(VO_2/HR), minute-ventilation (VE), ventilatory equivalents for oxygen and carbon dioxide(VE/VO_2 and VE/VCO_2) and tidal volume(VT). This allows for the assessment of ventilatory and cardiovascular response to MCT supplementation at rest.

Statistical Analysis

Jamovi software(2.6.44) was used to perform statistical analysis alongside Microsoft Excel which was used to store raw data. Before commencing analysis, they were plotted using boxplots and interquartile range (IQR) to screen for anomalies. If any data point lies $1.5 \times \text{IQR}$ below the lower quartile or above the upper quartile, it was deemed as an outlier and removed[21]. It was not possible to confirm if the data followed normal distribution since the sample size being small ($n=9$). Therefore, a non-parametric approach was taken. The Wilcoxon signed-rank test was used with statistical significance being set at $p < 0.05$ to examine the differences between pre- and post-supplementation for all the outcomes discussed earlier[22]. If a difference is identified, its effect size was calculated using rank-biserial correlation (r), with values of 0.1 being small, 0.3 being medium and 0.5 being large effects[23]. Medians and interquartile range are used to present the results for all outcomes. The results of the primary outcomes were visualised using boxplots and individual response plot to show both group and individual responses[24].

Results:

Data was collected from all nine participants and screened for anomalies using boxplot analysis as shown in figures 1-4. This was done to the primary outcomes – RER, RMR, VCO_2 and VO_2 . One participant (M6) was identified as an anomaly on RER, with their RER increasing post-ingestion of the supplement while all other participants showed a decrease (Figure 1). This could be due to improper mask seal during testing which was reported during testing resulting in a measurement error in breath-by-breath gas analysis. M6 was therefore excluded from data analysis. A summary of the anomaly detection results of the primary outcome differences is presented in table 3.

Table

Descriptives

	RER_Diff	RMR_diff	VCO2_diff	VO2_diff
Median	-0.0153	89.3	7.00e-4	0.0172
IQR	0.0331	740	0.0799	0.108
Minimum	-0.0523	-641	-0.0938	-0.0868
Maximum	0.0439	588	0.0719	0.0834

RER_Diff

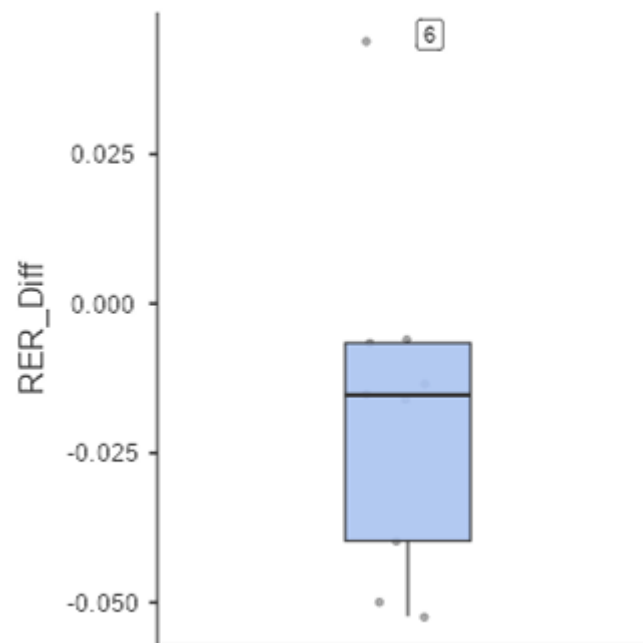


Figure – Boxplot of RER difference scores (post – pre) for all nine participants. M6 identified as anomaly (positive RER difference score)

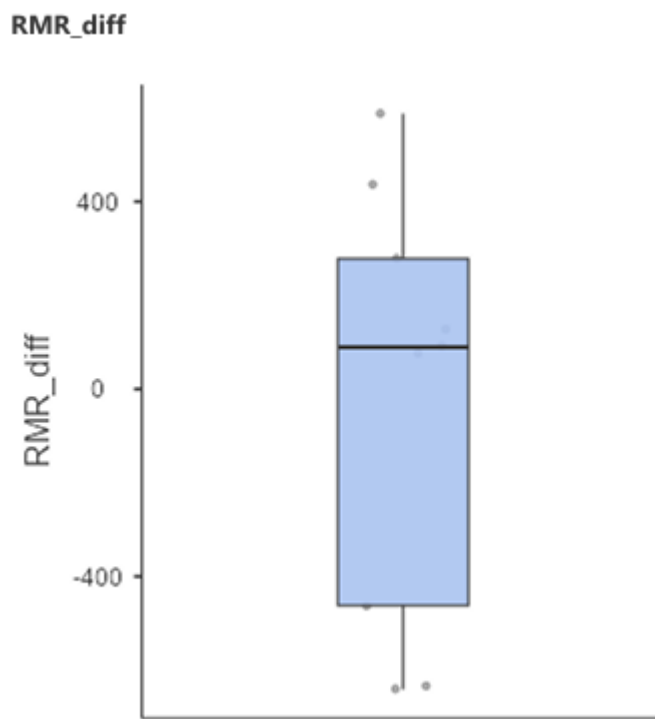


Figure – Boxplot of RMR difference scores (post – pre) for all nine participants. No anomalies identified

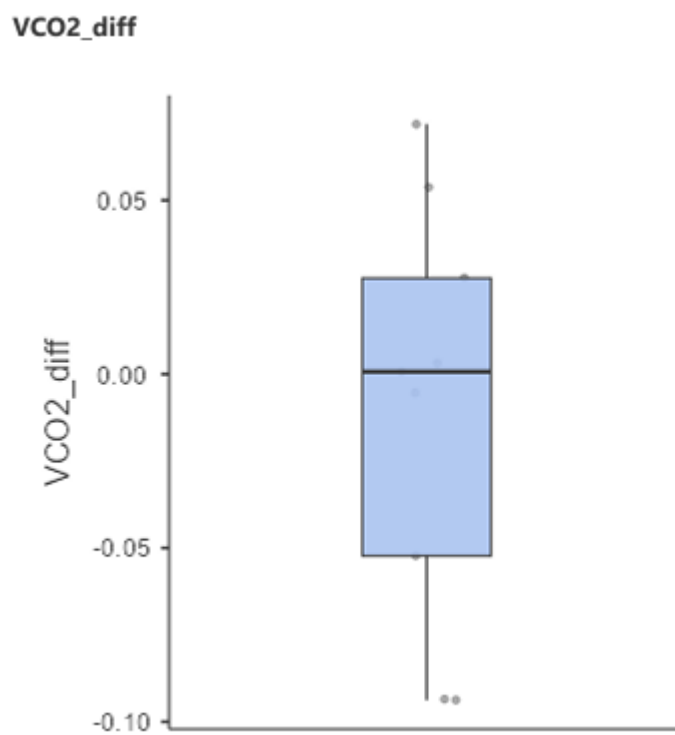


Figure - Boxplot of VCO₂ difference scores (post – pre) for all nine participants. No anomalies identified

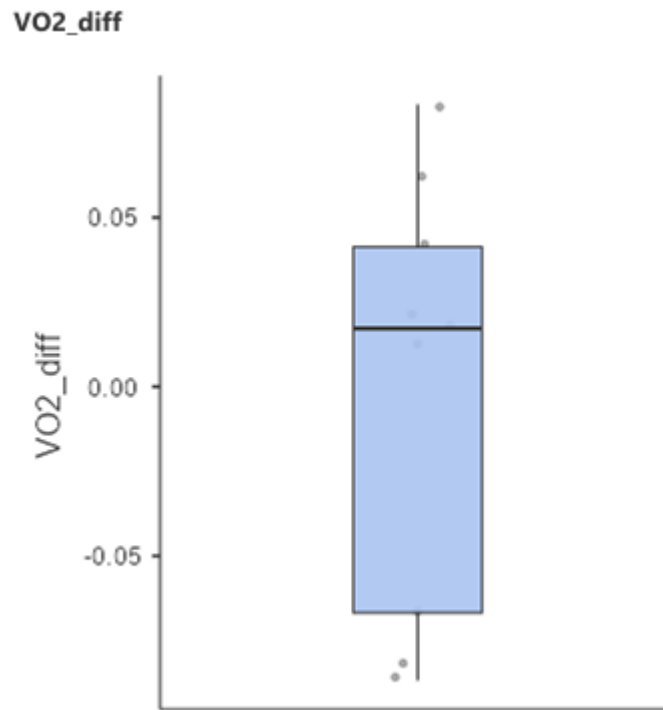


Figure - Boxplot of VO₂ difference scores (post – pre) for all nine participants. No anomalies identified
After exclusion of M6, data analysis was carried out for the remaining eight participants (5 male, 3 female). The updated participant characteristics is shown in table 4.

Table – Participant characteristics after anomaly exclusion

Descriptives					
	Sex	Height_cm	BMI	Weight_kg	Age
N	8	8	8	8	8
Mean		171	23.8	70.1	21.6
Standard deviation		8.22	3.86	16.0	0.518
Minimum		157	19.2	49.7	21
Maximum		183	29.6	89.5	22

Primary outcomes

Tables 5 and 6 show the results for the primary and secondary outcomes. Figures 5-8 illustrate the pre and post supplementation boxplots for the primary outcomes and Figures 9-12 show individual response plots.

RER showed a statistically significant decrease after ingestion of the MCT supplement ($W=36.00, p=0.008, r=1.000$). The median has decreased from 0.904[0.876-0.9333] pre-supplement to 0.882[0.863-0.894] post-supplement(Figures 5 and 9). The effect size being 1.000 indicates a perfect large effect, with all participants showing a decrease in RER post-supplementation(Figure 9). This indicates MCT consumption leads to greater fat oxidation.

RMR did not show a statistically significant change after MCT ingestion ($W=15.00$, $p=0.742$, $r=-0.167$). The median RMR showed very little change from 2408[1783-3181] kcal/day pre-supplement to 2430[1902-3239] kcal/day post-supplement (Figures 6 and 10). The small negative effect size indicates that the change is negligible meaning acute MCT consumption does not change RMR.

Similarly, both VO_2 and VCO_2 did not show statistically significant changes following MCT supplementation (VO_2 : $W=14.00$, $p=0.641$, $r=-0.222$; VCO_2 : $W=18.00$, $p=1.000$, $r=0.000$). The medians remained unchanged pre to post supplementation for both VO_2 (0.312[0.234-0.404] to 0.319[0.247-0.429] L/min) and VCO_2 (0.400[0.314-0.491] to 0.385[0.311-0.512] L/min) as shown in Figures 7,8,11 and 12.

Secondary outcomes:

None of the secondary outcome measures showed statistically significant changes after ingesting MCT. However, VE/VCO_2 had a large effect size with $r=-0.722$ despite not reaching the $p<0.05$ threshold. This could suggest MCT ingestion could trend towards increased ventilatory efficiency in eliminating carbon dioxide. The remaining secondary outcomes all showed non-significant changes suggesting no meaningful cardiovascular or ventilatory changes to acute MCT supplementation (Table 5 and 6).

Table – Wilcoxon signed-rank test results comparing pre and post MCT supplementation values for all primary and secondary outcome measures ($n=8$)

Paired Samples T-Test			Statistic	p		Effect Size
RER_pre	RER_post	Wilcoxon W	36.00	0.008	Rank biserial correlation	1.0000
RMR_day_pre	RMR_day_post	Wilcoxon W	15.00	0.742	Rank biserial correlation	-0.1667
VO2_pre	VO2_post	Wilcoxon W	14.00	0.641	Rank biserial correlation	-0.2222
VCO2_pre	VCO2_post	Wilcoxon W	18.00	1.000	Rank biserial correlation	0.0000
VO2kg_pre	VO2kg_post	Wilcoxon W	13.00	0.547	Rank biserial correlation	-0.2778
VO2HR_pre	VO2HR_post	Wilcoxon W	13.00	0.547	Rank biserial correlation	-0.2778
HR_pre	HR_post	Wilcoxon W	18.00	1.000	Rank biserial correlation	0.0000
VE_pre	VE_post	Wilcoxon W	12.00	0.461	Rank biserial correlation	-0.3333
VE_VO2_pre	VE_VO2_post	Wilcoxon W	11.00	0.383	Rank biserial correlation	-0.3889
VE_VCO2_pre	VE_VCO2_post	Wilcoxon W	5.00	0.078	Rank biserial correlation	-0.7222
VT_pre	VT_post	Wilcoxon W	17.00	0.945	Rank biserial correlation	-0.0556
BF_pre	BF_post	Wilcoxon W	14.00	0.641	Rank biserial correlation	-0.2222

Note. $H_2: \mu \text{ Measure 1} - \text{Measure 2} \neq 0$

Table – Descriptive statistics for all primary and secondary outcome measures pre and post MCT supplementation ($n=8$)

Descriptives

	N	Mean	Median	SD	SE
RER_pre	8	0.903	0.904	0.0311	0.01101
RER_post	8	0.878	0.882	0.0187	0.00661
RMR_day_pre	8	2427.363	2408.000	658.0218	232.64583
RMR_day_post	8	2468.250	2430.250	543.9688	192.32200
VO2_pre	8	0.316	0.312	0.0920	0.03254
VO2_post	8	0.324	0.319	0.0747	0.02641
VCO2_pre	8	0.400	0.400	0.0856	0.03027
VCO2_post	8	0.395	0.385	0.0764	0.02702
VO2kg_pre	8	4.529	4.419	0.9912	0.35045
VO2kg_post	8	4.719	4.979	0.9237	0.32658
VO2HR_pre	8	4.499	4.280	1.0032	0.35467
VO2HR_post	8	4.598	4.297	0.8634	0.30526
HR_pre	8	69.610	66.114	7.8818	2.78663
HR_post	8	70.312	72.448	7.1507	2.52815
VE_pre	8	9.824	10.077	2.2526	0.79641
VE_post	8	10.270	9.960	1.8525	0.65497
VE_VO2_pre	8	22.082	22.319	1.6566	0.58569
VE_VO2_post	8	22.934	22.274	2.2355	0.79036
VE_VCO2_pre	8	24.426	24.697	1.7662	0.62445
VE_VCO2_post	8	26.070	25.239	2.6250	0.92810
VT_pre	8	0.571	0.547	0.1543	0.05456
VT_post	8	0.561	0.555	0.0850	0.03004
BF_pre	8	18.497	18.592	2.1200	0.74953
BF_post	8	19.043	18.773	2.3542	0.83233

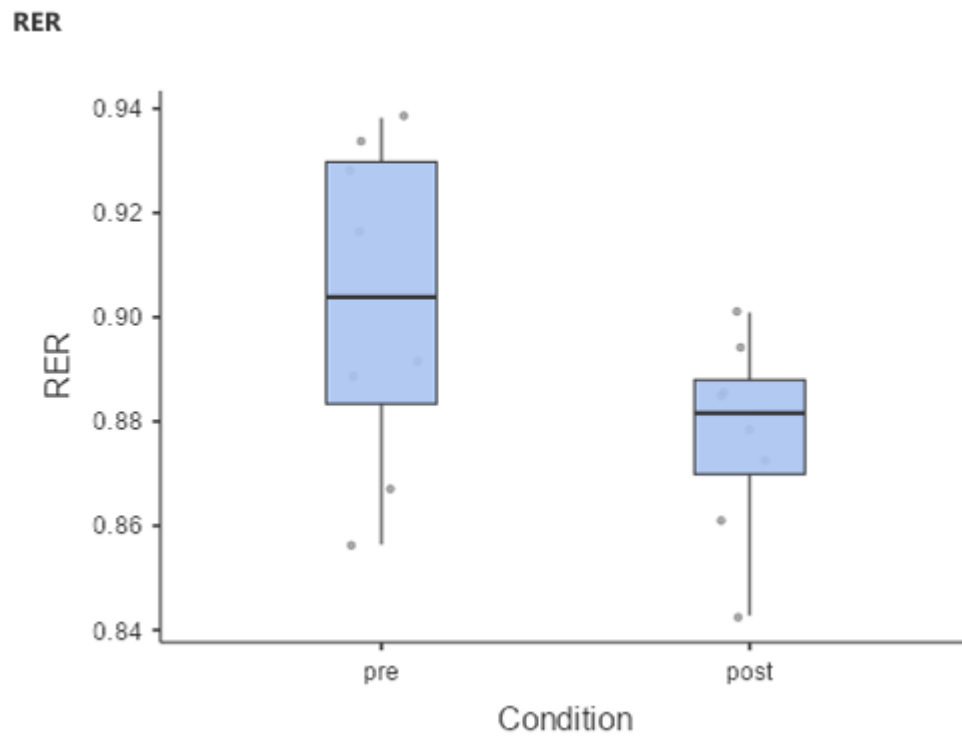


Figure - Boxplot showing RER pre and post MCT supplementation (n=8). Horizontal line represents median; whiskers represent $1.5 \times \text{IQR}$; dots represent individual data points.

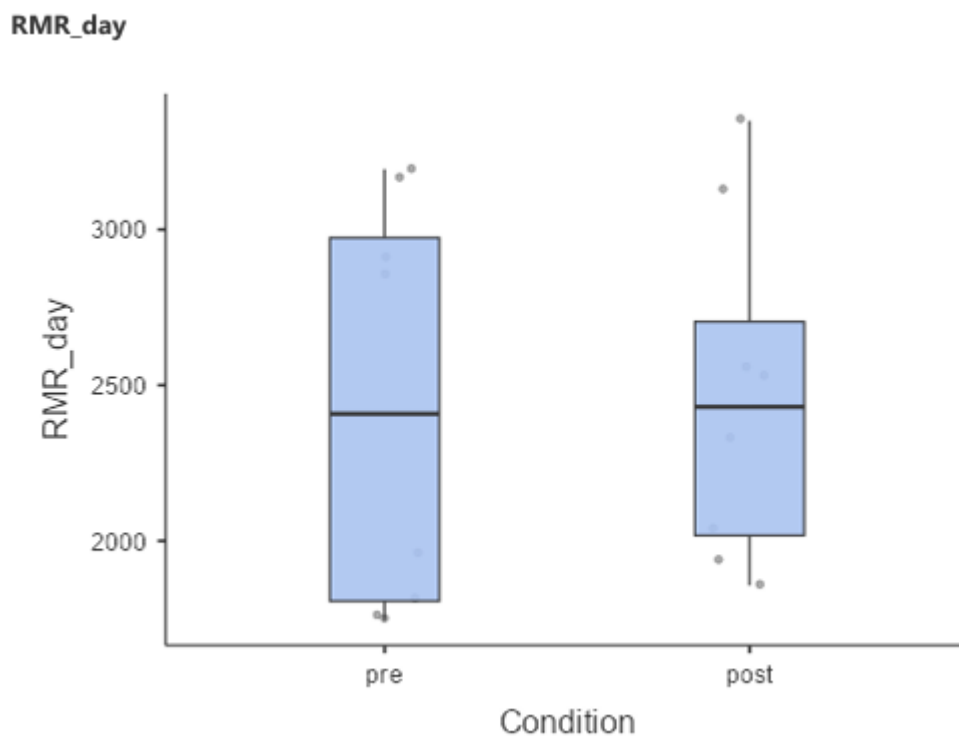


Figure - Boxplot showing RMR (kcal/day) pre and post MCT supplementation (n=8). Horizontal line represents median; whiskers represent $1.5 \times \text{IQR}$; dots represent individual data points.

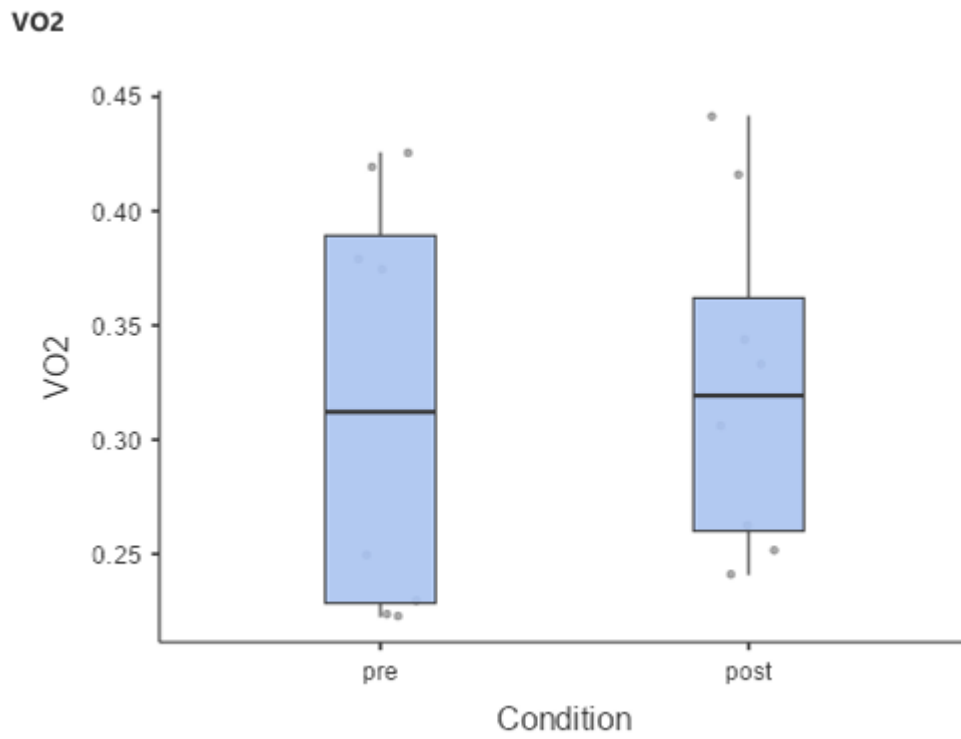


Figure - Boxplot showing $\dot{V}O_2$ (L/min) pre and post MCT supplementation (n=8). Horizontal line represents median; whiskers represent $1.5 \times \text{IQR}$; dots represent individual data points.

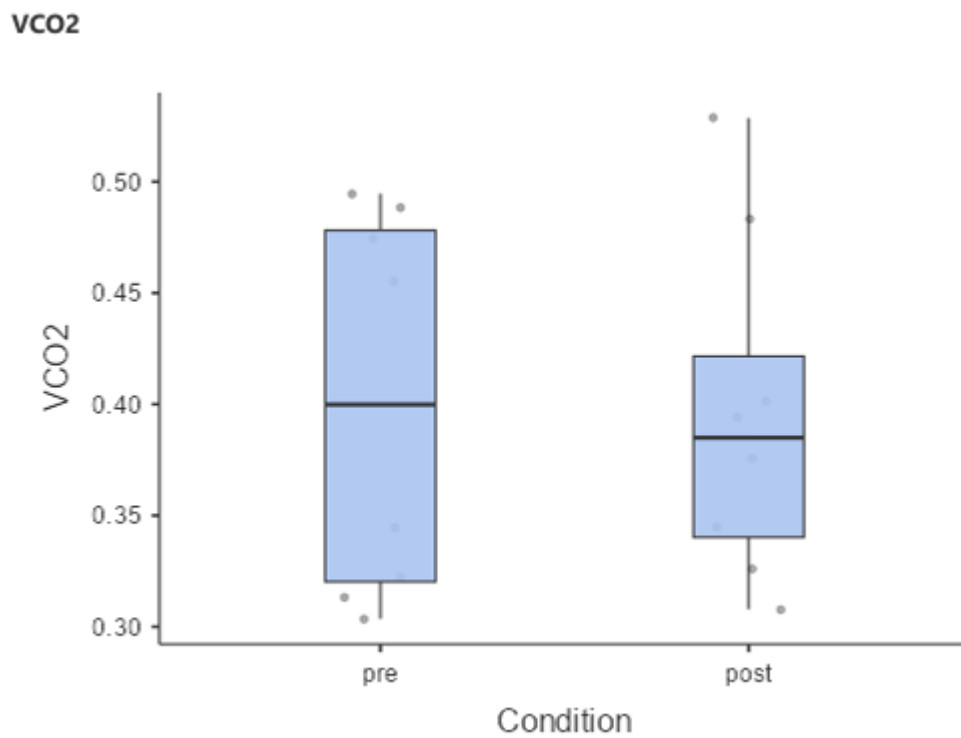


Figure - Boxplot showing $\dot{V}CO_2$ (L/min) pre and post MCT supplementation (n=8). Horizontal line represents median; whiskers represent $1.5 \times \text{IQR}$; dots represent individual data points.

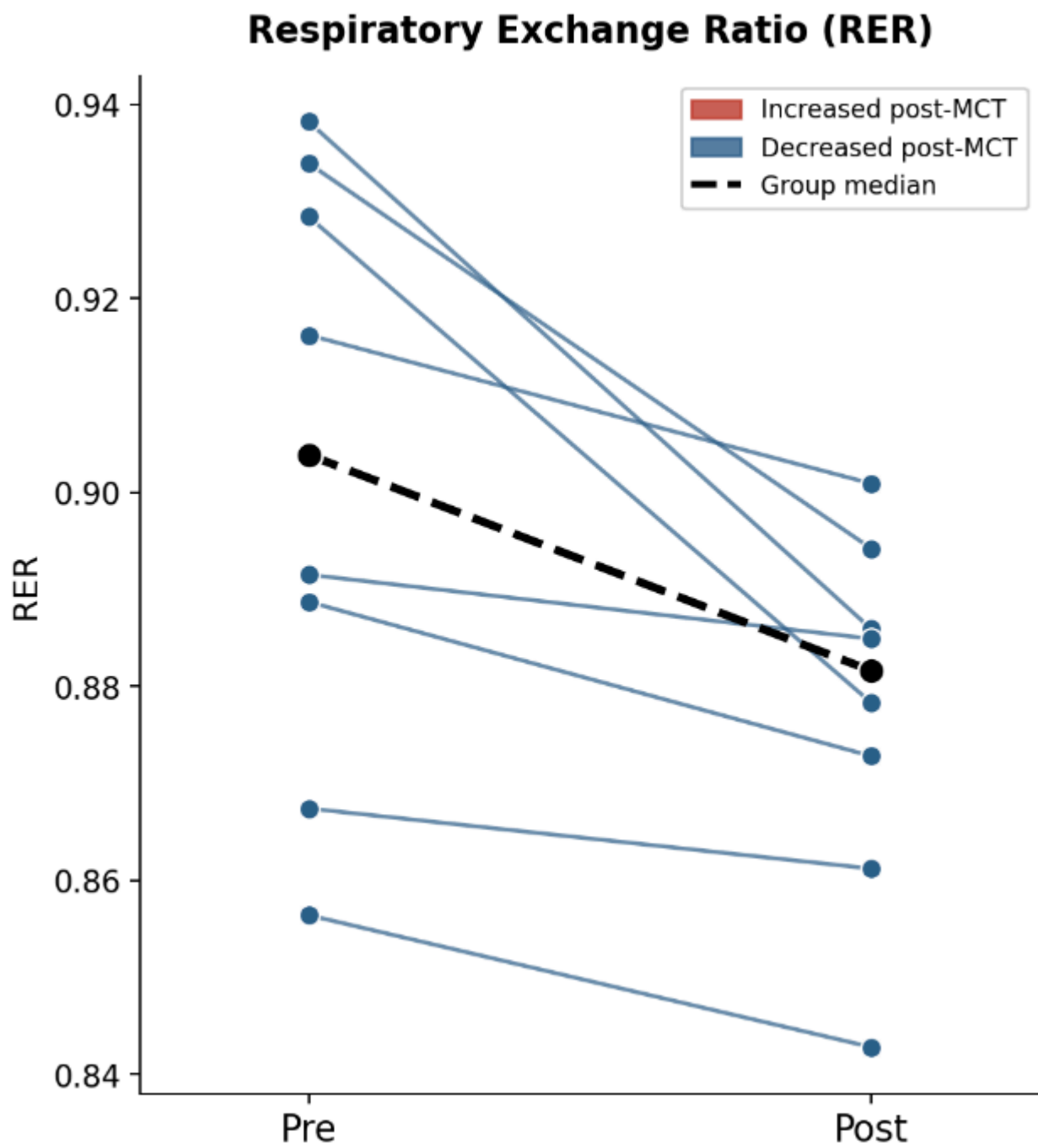


Figure - Individual response plot showing RER for each participant pre and post MCT supplementation (n=8). All eight participants demonstrated a decrease in RER post-supplementation.

Resting Metabolic Rate (RMR)

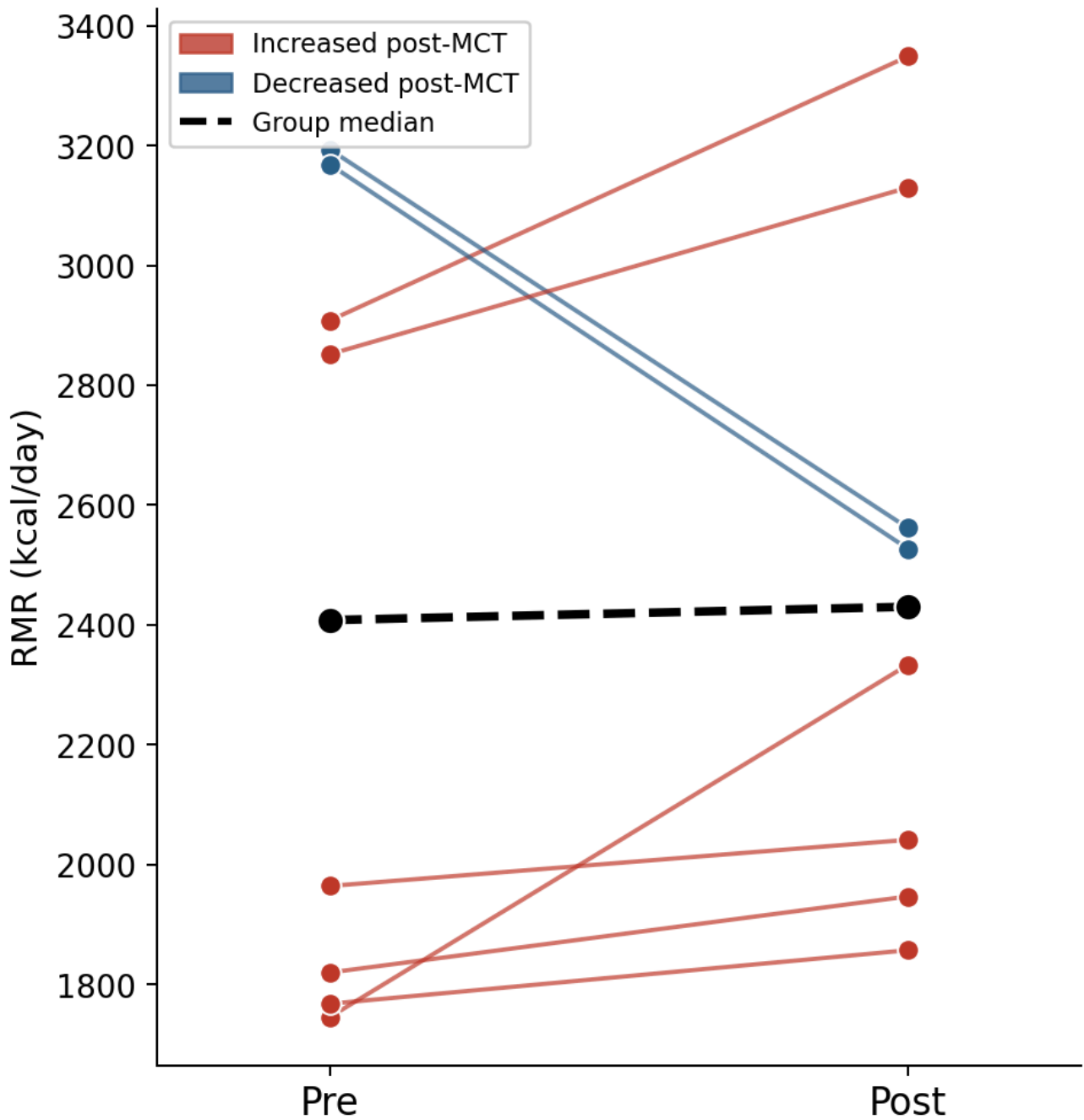


Figure - Individual response plot showing RMR (kcal/day) for each participant pre and post MCT supplementation (n=8).

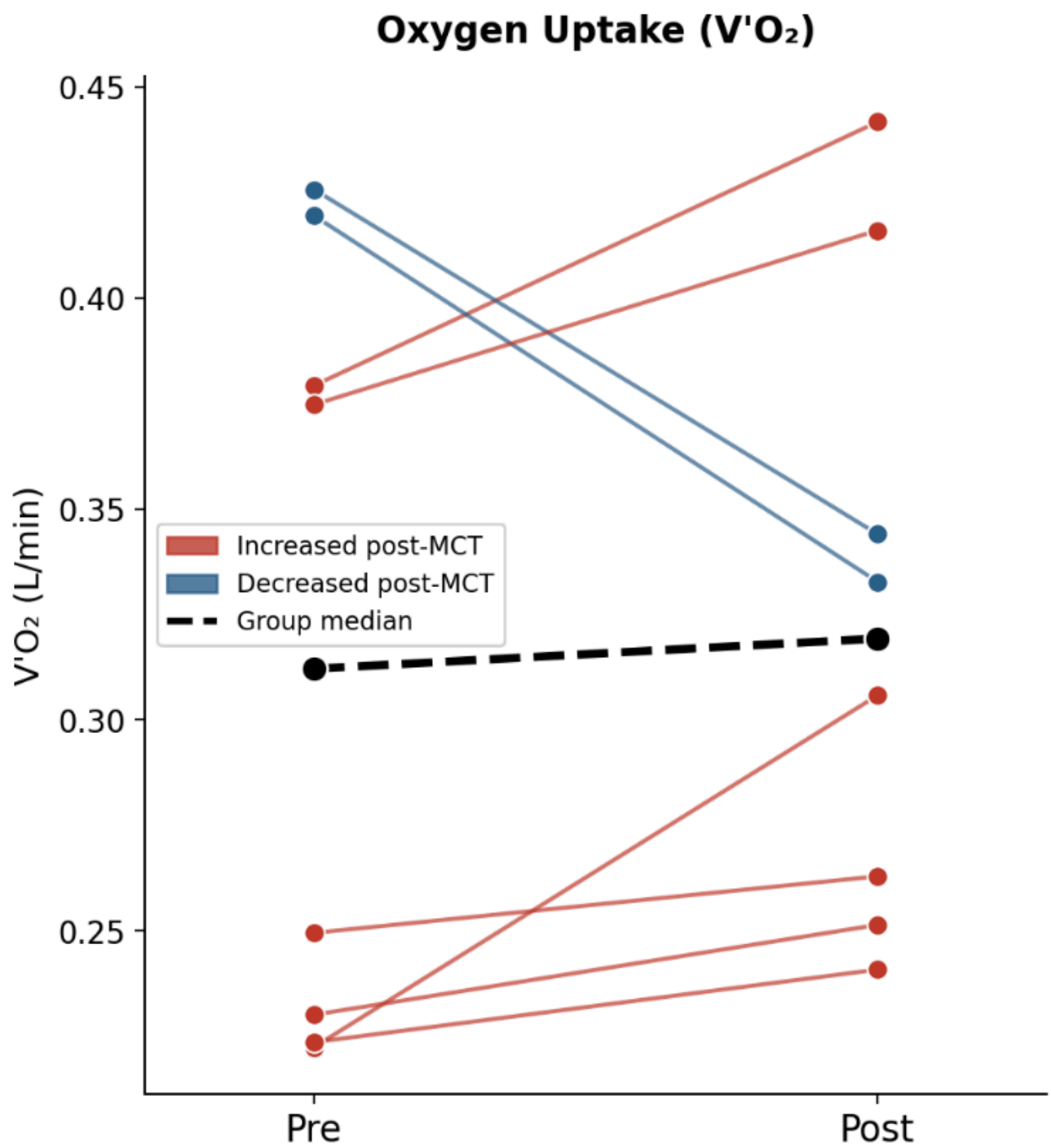


Figure - Individual response plot showing $\dot{V}O_2$ (L/min) for each participant pre and post MCT supplementation (n=8).

Carbon Dioxide Production ($\dot{V}\text{CO}_2$)

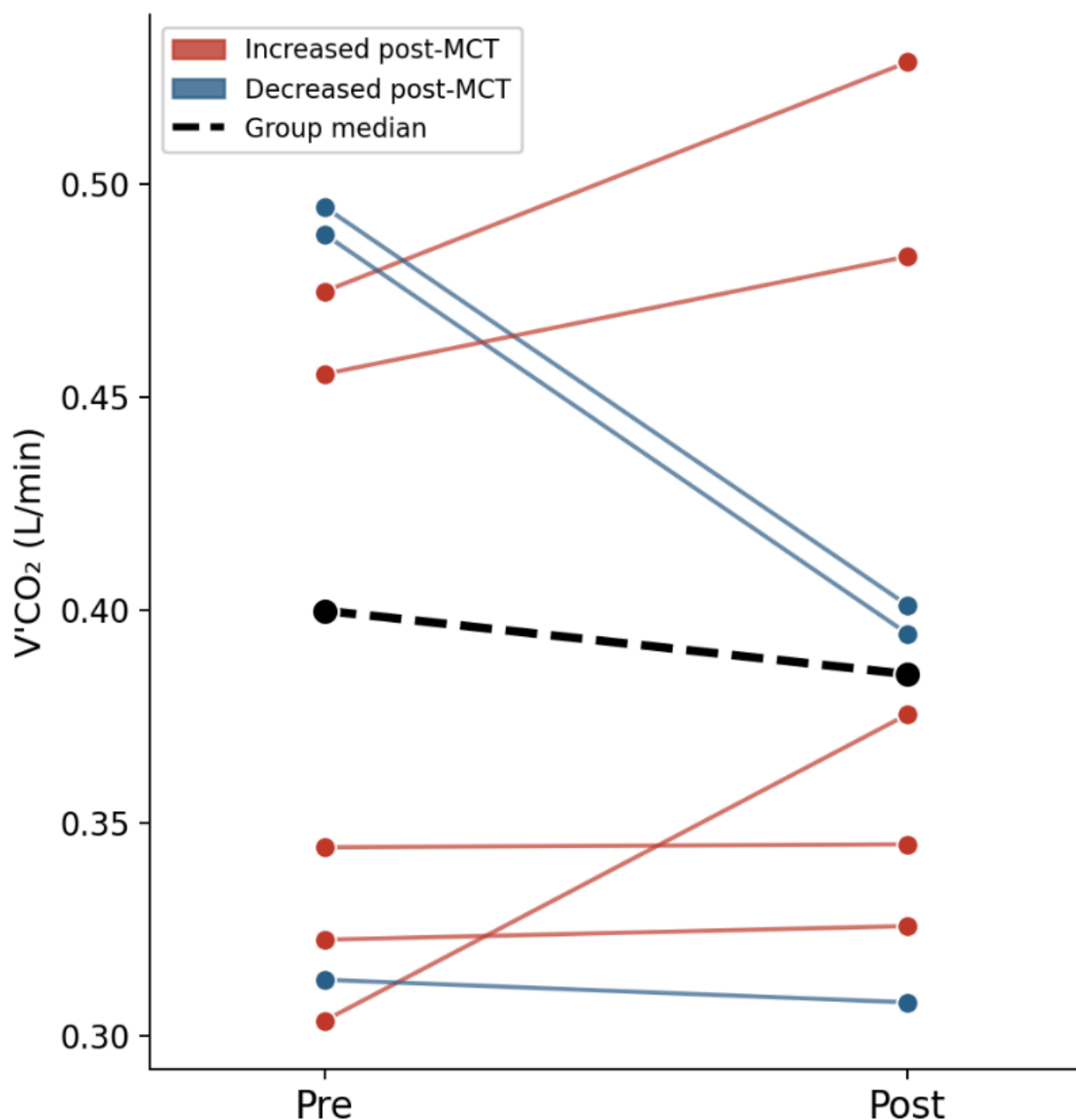


Figure - Individual response plot showing $\dot{V}\text{CO}_2$ (L/min) for each participant pre and post MCT supplementation (n=8).

Discussion:

This research study assessed the acute effects of a single 30g C8 MCT oil supplement on RMR and fat oxidation at rest in young healthy adults, measured three hours post-ingestion using breath-by-breath gas analysis. The main finding was a significant decrease in RER post supplementation, suggesting a shift toward greater fat oxidation at rest. However, RMR did not change significantly, and other secondary outcome measures showed no significant differences with the exception of $\text{VE}/\dot{V}\text{CO}_2$ being close to achieving significance ($p=0.078$).

Primary outcomes

MCT supplementation was hypothesised to shift toward greater fat oxidation at rest. The significant reduction in RER supports this. RER is an excellent indicator of substrate oxidation; values over 1.00 indicates reliance on carbohydrate oxidation while values closer to 0.70 indicates predominant fat oxidation[25]. The decrease from 0.904 to 0.882 is a meaningful shift towards fat oxidation following MCT ingestion. This is further reinforced by the perfect effect size of 1.000 where all 8 participants showed this consistent decrease in RER following MCT consumption.

The metabolic properties of C8 MCTs are likely the reason for these results. C8 MCFAs cross the mitochondrial membrane without the need for the CPT-I shuttle system allowing for faster transport and oxidation[4]. This likely led to the increase in overall fat oxidation and thus reducing RER. Pure C8 MCT oil was used in this study which may have enhanced the effect compared to existing studies that used a mix of C8 and C10 MCT oil. C8 is more rapidly oxidised than C10[26]. Overall, these findings are quite consistent with other studies such as Papamandjaris et al.(1998) and St-Onge& Jones(2002) who reported similar increases in RER[4][11].

However, RMR did not have statistically significant changes after MCT consumption. The median RMR increased by 22kcal/day which is negligible. Some of the existing evidence reported increased thermogenesis post MCT ingestion which is inconsistent with the findings from this study[11][17]. This increased thermogenesis could be observed if the MCT supplementation was given in a long-term supplementation regime instead of a single acute dose[27].

Secondary outcomes

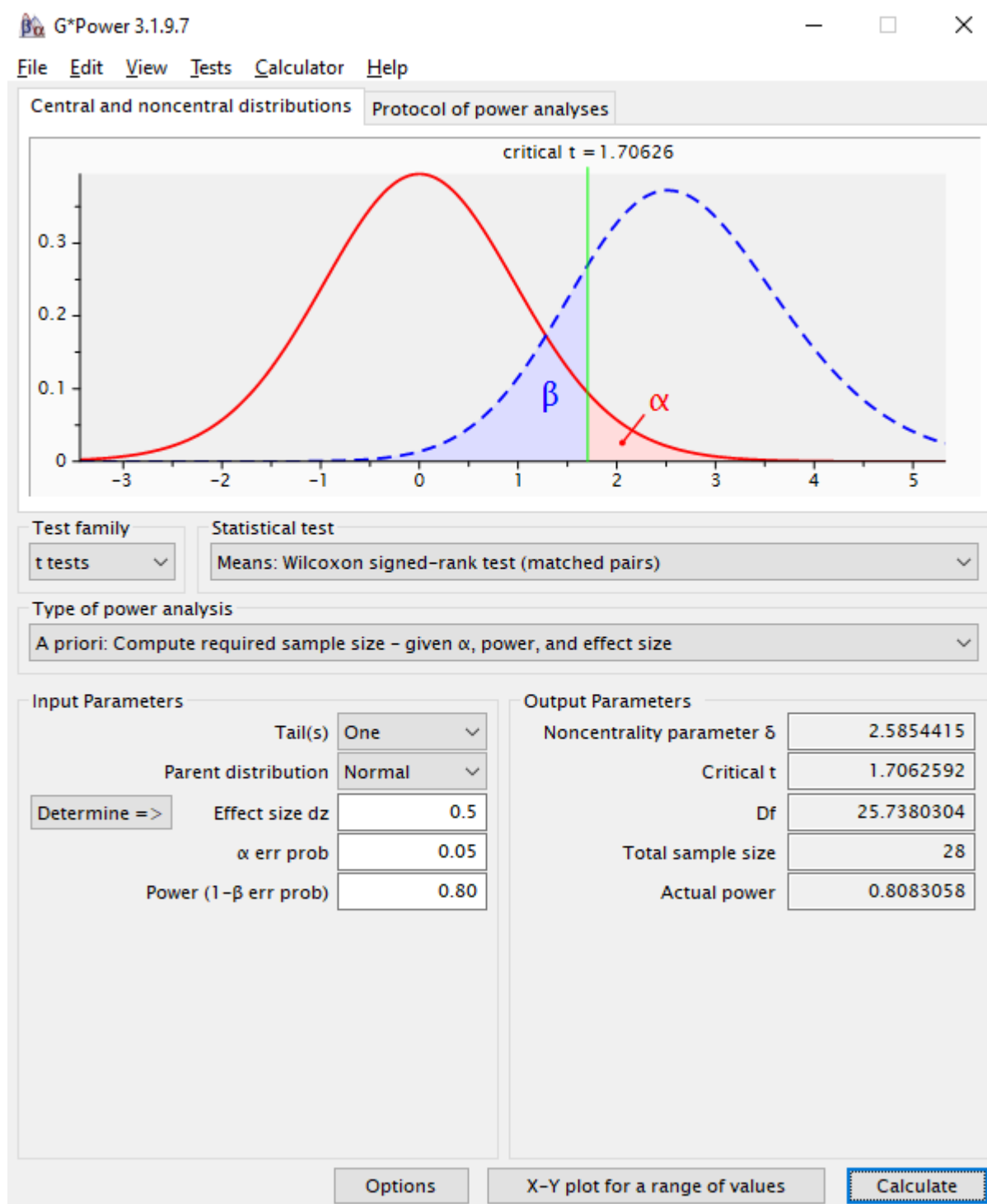
With regard to the secondary outcomes, VE/VCO_2 approached significance showing a large effect size despite not being statistically significant by reaching $p < 0.05$. VE/VCO_2 represents ventilatory efficiency in removing carbon dioxide. A reduction in this suggests that MCT supplementation could improve ventilation efficiency by potentially affecting metabolic gas exchange patterns related with increased fat oxidation[28]. This is consistent with the meaningful decrease in RER from earlier. Greater fat oxidation produces less CO_2 per unit of oxygen consumed making its removal more efficient[29].

However, all other secondary outcome measures did not show any meaningful changes with small effect sizes ($r < 0.3$). This indicates acute MCT ingestion does not affect cardiovascular or respiratory function significantly at rest. Some studies have reported acute MCT ingestion has noticeable effects on the sympathetic nervous system leading to increased resting heart rate (HR)[30]. However, the results from this study does not show any meaningful changes in HR. This may be due to lowered sympathetic response to dietary fats in healthy recreationally active individuals[31].

Limitations

There are several limitations in the study. The most significant size being the sample size of 9 participants which was reduced to 8 after anomaly screening. This reduced statistical power greatly[32]. A priori power analysis was carried out when the study was designed indicating a minimum of 28 participants were needed to detect a significant effect (Figure 13)[33]. This study achieved 29% of the target due to time constraints. This increases the likelihood of type 2 error – limiting the applicability of findings[34]. This

could explain the non-significant findings seen in RMR and the secondary outcome measures despite some of them showing medium to large effect sizes.



After the first testing, participants were able to leave the laboratory for three hours unsupervised being instructed to avoid food and exercise. It is not possible to confirm compliance and thus this could have confounded post-supplementation measurements. In addition to this, pre-supplementation testing was conducted between 08:00-10:00 and post-supplementation testing between 11:00-14:00. Circadian variation in RMR and substrate oxidation could have affected the metabolic measurements taken making it not possible to dissociate the effects of MCT from natural circadian variation[35].

The study did not have a control condition such as long-chain triglycerides or isocaloric placebos – something commonly seen in existing literature. Therefore, one cannot attribute the RER decrease to MCT specifically. The macronutrients and calories in the mixture from the cacao powder and milk itself could have had an effect on metabolism[36].

Future studies should have appropriate controls in order to isolate the effects of MCT supplementation alone. This can be done by using randomised crossover designs with isocaloric LCT comparators. Larger sample sizes should be used allowing for statistical significance. Furthermore, to avoid the effects of circadian variation on metabolic variables, the pre and post supplement test could be done on separate days at the same time.

Conclusion

Acute ingestion of 30g of C8 MCT oil has led to a statistically significant decrease in RER in young healthy adults three hours post-consumption. This illustrates a change in substrate utilisation and a shift towards using fat oxidation for energy across all participants. However, RMR and other secondary outcome measures did not change significantly. These results support the hypothesis partially that acute MCT supplementation increases fat oxidation at rest. It also suggests that one acute consumption at 30g does not lead to notable changes in RMR. The sample size, coupled with the lack of control condition has limited the study. Future research needs to have larger crossover designs with controls in order to better understand the acute metabolic effects of MCT supplementation in young healthy adults.

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